

Algorithms for Data Science

Lecture 1:

Introduction to the Course and The Data Science Pipeline



Grading Schema

- Midterm: 15 Degree
- Activities
 - Project: 20 Degree
 - Quizzes (LMS): 20 Degree (5 quizzes, Inside lab, best 4)
 - Practical Assignments: 20 Degree (LMS + Discussion)
- Final Exam: 25 Degree



What is this course about?

- **Course Title:** Algorithms for Data Science
Prerequisite: Data Structure and Algorithms
- This course is **not** just about learning to code. It is about the **computational techniques** and **algorithmic thinking** required to extract meaningful insights from complex datasets.
- **What this course is not:**
 - Not a general programming course
 - Not solely a machine learning theory course
 - Not a big data infrastructure course

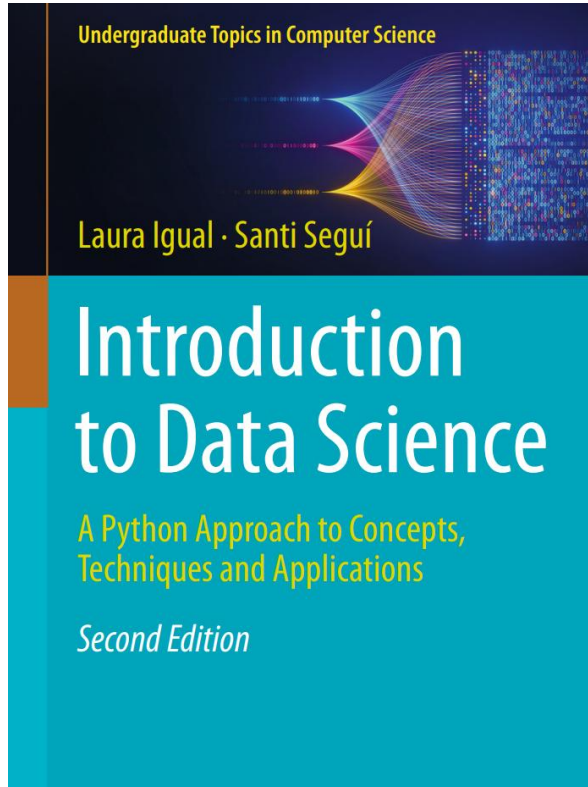


What is this course about?

- **This Course is about:** How do we take raw data and turn it into a decision using algorithms?
- In CET344, we focus on the **algorithms** that used in:
 - **Data Preprocessing** (Handling data)
 - **Feature Engineering** (Selecting the right variables)
 - **Classification** (Predicting categories)
 - **Regression** (Forecasting numbers)
 - **Clustering** (Grouping unlabeled data)



Reference Books (1/2)



This book will be uploaded to the LMS



Reference Books (1/2)



Data Science

A First Introduction with Python

Tiffany Timbers, Trevor Campbell, Melissa Lee, Joel Ostblom, Lindsey Heagy

The link to this online book:

[https://python.datasciencebook.ca/
index.html](https://python.datasciencebook.ca/index.html)





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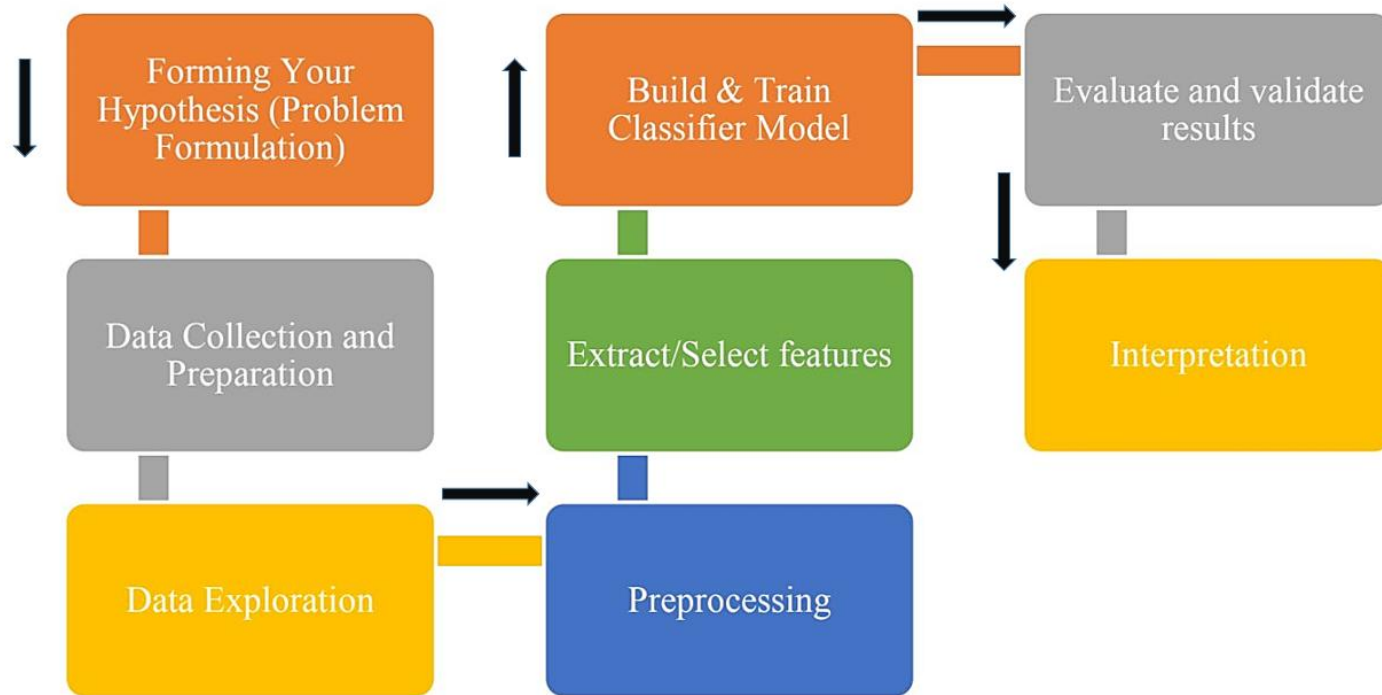
- **Overview about the Data Science Pipeline**

- ✓ Phase 1: Forming Your Hypothesis
- ✓ Phase 2: Data Collection & Preparation
- ✓ Phase 3: Data Exploration
- ✓ Phase 4: Preprocessing
- ✓ Phase 5: Feature Extraction & Selection
- ✓ Phase 6: Build & Train Classifier Model
- ✓ Phase 7: Evaluate & Validate Results
- ✓ Phase 8: Interpretation



The Data Science Pipeline

In this course, we follow a structured pipeline:



Phase 1: Forming Your Hypothesis (Problem Formulation)

- Every data science project begins with a **question**, not data.
- Data does not speak for itself. A million rows of numbers, text, or images contain infinite possible stories. Without a question, you cannot:
 - Choose which algorithm to use
 - Decide which features matter
 - Know if your results are meaningful
 - Determine if the project succeeded or failed
- **The question is the compass. The algorithms are the vehicle. The data is the terrain.**



Phase 1: Forming Your Hypothesis (Problem Formulation)

- **What is a Hypothesis in Data Science?**

A hypothesis is a testable statement that transforms your open-ended question into a focused, algorithm-ready investigation.

Your Question	Your Hypothesis (Testable Statement)
"Can we predict which customers will leave?"	Relationship: "Customers with more than 3 support calls in 30 days are 4x more likely to churn."
"What drives housing prices?"	Relationship: "Square footage and location are the two strongest predictors of sale price."
"Are there groups in this unlabeled data?"	Structure: "This dataset can be partitioned into meaningful clusters based on purchase history."



Phase 1: Forming Your Hypothesis (Problem Formulation)

- Without a hypothesis, you cannot choose an algorithm.

Hypothesis Example	Problem Type	Algorithm Category
"Transaction X is fraudulent or legitimate"	Classification	Supervised (Discrete output)
"Housing price Y is a function of square footage, location, age"	Regression	Supervised (Continuous output)
"Customers naturally group into 3–5 distinct segments"	Clustering	Unsupervised
"Frequently purchased items occur together"	Pattern Discovery	Association rule mining



Phase 2: Data Collection and Preparation

- Once the hypothesis is formed, we acquire the data needed to test it.
- **Data Sources**

Source Type	Examples
Internal	Company databases, transaction logs, sensor readings
External	Open data portals, government databases, APIs
Generated	Surveys, experiments, simulations



Phase 2: Data Collection and Preparation

- **Data Preparation Tasks**

- **Loading** – Reading data from files, databases, or streams
- **Integration** – Combining data from multiple sources
- **Unit Conversion** – Ensuring consistent measurement scales
- **Storage** – Organizing data for efficient access

- **Challenge:** Real-world data is rarely ready for immediate analysis. Preparation is not optional.



Phase 3: Data Exploration

- Before cleaning or modeling, we must **understand** the data.
- **Goals of Data Exploration**
 - Identify structure and patterns
 - Detect anomalies and errors
 - Formulate new hypotheses
 - Guide preprocessing and feature selection



Phase 3: Data Exploration

- **Exploratory Algorithms (Descriptive Statistics)**
- **Measures of Central Tendency**

Measure	Definition	Use Case
Mean	$\text{Sum} \div N$	Symmetric, continuous data
Median	Middle value	Skewed distributions, robust to outliers
Mode	Most frequent value	Categorical data



Phase 3: Data Exploration

- **Example: Understanding Distribution**

A dataset of customer ages:

- Mean = 52 years
- Median = 47 years

- **Interpretation:** Positive skew. Older outliers pull the mean upward.

- **Why this matters for algorithms:**

- Algorithms using distance metrics (K-Means, K-NN) assume roughly symmetric distributions.
- Skewed data may require transformation **before** preprocessing.



Phase 4: Preprocessing

- Preprocessing algorithms transform raw data into a **clean, consistent state** suitable for modeling.
- **Core Preprocessing Tasks**

Task	Problem	Algorithmic Approach
Missing Values	Incomplete records	Imputation (mean, median, K-NN), deletion
Noise Reduction	Measurement errors, outliers	Smoothing, binning, thresholding
Scaling	Features on different units	Normalization (0–1), Standardization (z-score)

Phase 4: Preprocessing

- **Why Preprocessing is Algorithmic**

- Preprocessing is not arbitrary "cleanup." It involves **decisions**:
- Should we delete a row with missing data or estimate the value?
- What imputation method preserves the underlying distribution?
- Does scaling change the interpretation of the model?



Phase 5: Feature Extraction and Selection

- Features are the individual inputs an algorithm uses to decide. In a spreadsheet, features are the columns. In an image, features are the pixels.

Task	Definition	Example
Feature Extraction	Creating new features from raw data	Pixel intensities → edge detection
Feature Construction	Combining existing features	Ratio of length to width
Feature Selection	Choosing the most relevant subset	Eliminating redundant measurements
Feature Reduction	Reducing the number of features while preserving important information	PCA (Principal Component Analysis)

Phase 5: Feature Extraction and Selection

- **The Curse of Dimensionality**

As the number of features increases:

- Data becomes sparse
- Distance metrics become less meaningful
- Computational cost increases
- Risk of overfitting increases

- **Solution:** Dimensionality reduction algorithms (PCA, t-SNE) and feature selection algorithms.



Phase 6: Build & Train Classifier Model

- This phase contains the algorithms most commonly associated with "data science."
- **Taxonomy of Modeling Algorithms**

Task	Algorithm Examples
Classification	Decision Trees, k-Nearest Neighbors, SVM, Logistic Regression
Regression	Linear Regression, Ridge, Lasso, Polynomial Regression
Clustering	K-Means, DBSCAN, Hierarchical
Dimensionality Reduction	PCA, t-SNE



Phase 6: Build & Train Classifier Model

- **The Training Process**

- Split data into **training** and **test** sets
- Present training data to algorithm
- Algorithm adjusts internal parameters to minimize error
- Result: A **model** (decision boundary, regression line, cluster assignments)



Phase 7: Evaluate and Validate Results

- A model is only useful if it **generalizes** to new, unseen data.
- **The Fundamental Problem**
 - ✓ **Overfitting:** The model memorizes the training data, including its noise.
 - ✓ **Underfitting:** The model fails to capture the underlying pattern.
- The test set is used to evaluate the model.



Phase 7: Evaluate and Validate Results

- Many Evaluation metrics can be used to evaluate the model's performance.
The evaluation metrics used based on the task.

Task	Metrics
Classification	Accuracy, Precision, Recall, F1-Score, ROC-AUC
Regression	MSE, RMSE, MAE, R^2



Phase 8: Interpretation

- The final phase: translating model outputs into **decisions and insights**.
- **Interpretation Questions**
 - What features were most important to the model?
 - Can we explain a single prediction?
 - What action should be taken based on this result?



Phase 8: Interpretation

- **Interpretability vs. Performance Tradeoff**

Model Type	Interpretability	Example
High Interpretability	High	Decision Trees, Linear Regression
Low Interpretability	Low	Random Forests, Neural Networks

- We prioritize algorithms that balance performance with interpretability for data science practice.



Any Question?

